

Tusher Bhomik

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Kaggle](#)

EDUCATION

• Bangladesh University of Engineering and Technology

Bachelor of Science in CSE

2022 – Present (currently at 8th semester)

Dhaka, Bangladesh

PROJECTS

• HealthSync

Spring Boot, React, PostgreSQL, Firebase

2025

[GitHub](#)

- Developed a comprehensive healthcare platform connecting patients and doctors with appointment scheduling, digital prescription management, and medical history tracking.

• Arxiv Research Paper RAG System

Python, SQL, SQLite, FastAPI, ChromaDB, Data Analysis

2026

[GitHub](#)

- Built an end-to-end **data pipeline** for ingesting, cleaning, and analyzing arXiv research paper data using Python and SQL.
- Developed a **RAG-based API system** using FastAPI and ChromaDB for semantic retrieval and question answering.
- Generated analytical insights with **visualizations** and implemented automated benchmarking using query-based evaluation.

• Mini C Compiler

CSE-310 Compiler Design, C, 8086 Assembly

2024

[GitHub](#)

- Built a **mini C compiler** from scratch in four phases: Symbol Table, Lexical Analysis, Semantic Analysis, and Intermediate Code Generation.
- Generated **8086 assembly** as intermediate output to validate end-to-end compilation workflow.

• Enhanced xv6-riscv Kernel

C, Assembly, Makefile

2024

[GitHub](#)

- Implemented an **MLFQ scheduler** with aging and added **kernel-level threading** support.
- Added custom system calls (trace, sysinfo) and a Bash autograder for testing.

• Dealytics

Marketplace Analytics Platform

2025

[GitHub](#)

- Building Bangladesh's first **analytics-driven marketplace aggregator** to track real-time price trends and predict discounts across markets.
- Implementing **data-driven inventory optimization** for sellers and a secure negotiation system to facilitate transparent trading.

RESEARCH

• Fall Risk Estimation from Gait using Deep Learning (Ongoing Thesis)

Supervisor: Mahmuda Naznin, Dept. of CSE, BUET

2025 – Present

Dhaka, Bangladesh

- Developing an interpretable deep learning framework to estimate fall vulnerability from motion capture gait data by detecting joint-level deviations and inferring biomechanical causes of fall risk.

• Improving Function Calling in Large Language Models using Reinforcement Learning (Ongoing)

Supervisor: Md. Rizwan Parvez, QCRI Research Institute

2026 – Present

Doha, Qatar

- Investigating reinforcement learning-based optimization to enhance the reliability and accuracy of function calling behavior in large language models for complex tool-use scenarios.

SKILLS

- **Programming:** Python, C++, C, Java, JavaScript | **Databases:** MySQL, PostgreSQL, MongoDB
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, LLMs, vLLM
- **Tools:** Git, Docker, Linux | **Frontend:** React, Next.js, Material-UI

RELEVANT COURSEWORK

- **Relevant Coursework** Data Structure, Algorithm, Computer Networks, Database Management System, Compiler Design, Operating System, Software Engineering, Artificial Intelligence, Machine Learning, Computer Architecture, Theory of Computation, Computer Graphics

ADDITIONAL INFORMATION

Languages: English, Bengali. **Interests:** Photography, Traveling, Badminton, Cycling.